Sample Preparation for 10x Genomics

A How To Guide: Considerations and Best Practices

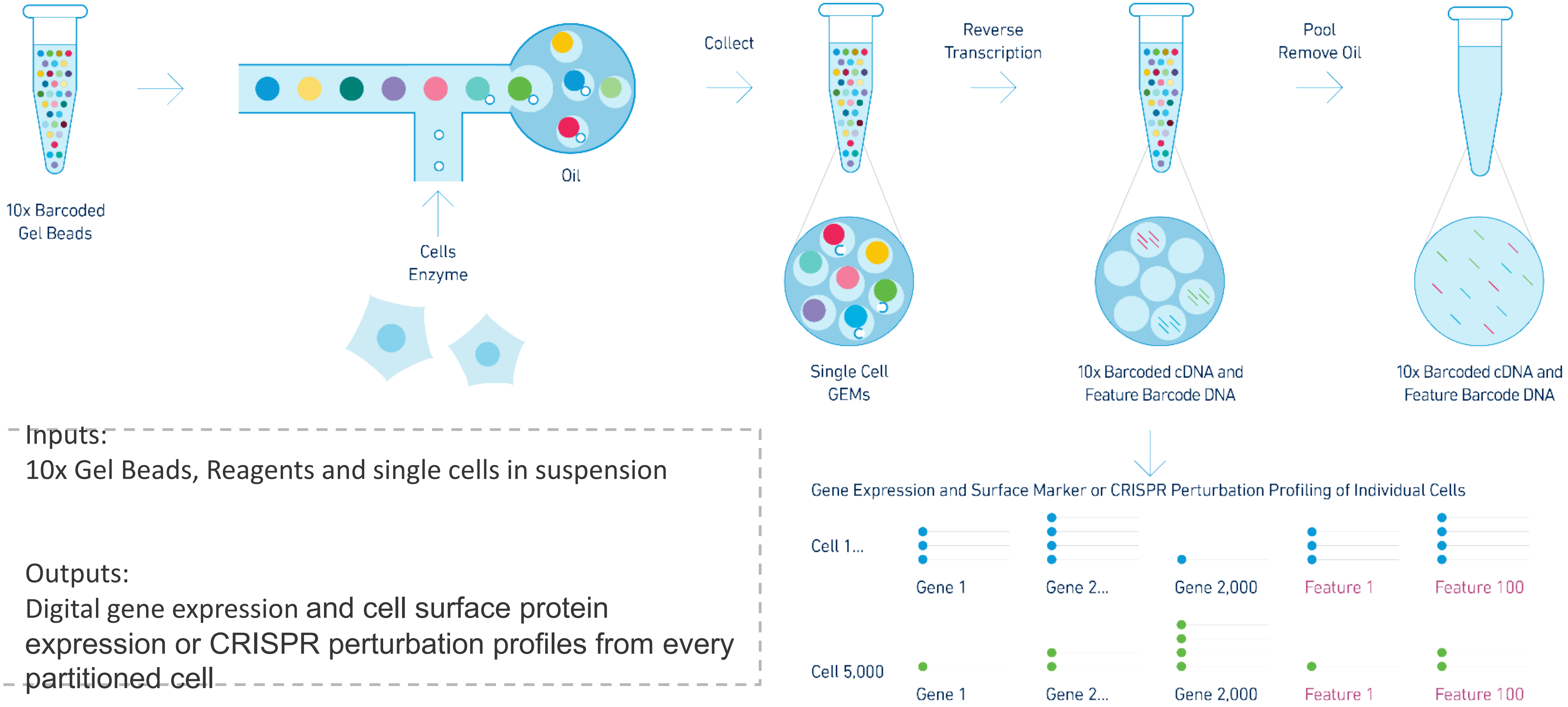
Bashir Sadet, PhD
Senior Field Application Scientist

Variables important to the success of single cell experiments

- **Objective of the talk:**
  - Identify variables that are crucial for a successful single cell RNA/ATAC-seq experiment
  - +/- feature barcoding
- **Important variables that will be discussed:**
  - Viability
  - Counting
  - Clumps and debris
  - Absence of inhibitors of enzymatic reactions (EDTA, Mg++, Ca++)
  - Optimization of protocols beforehand

Chromium Single Cell Gene Expression Solution with Feature Barcoding technology

Workflow Overview

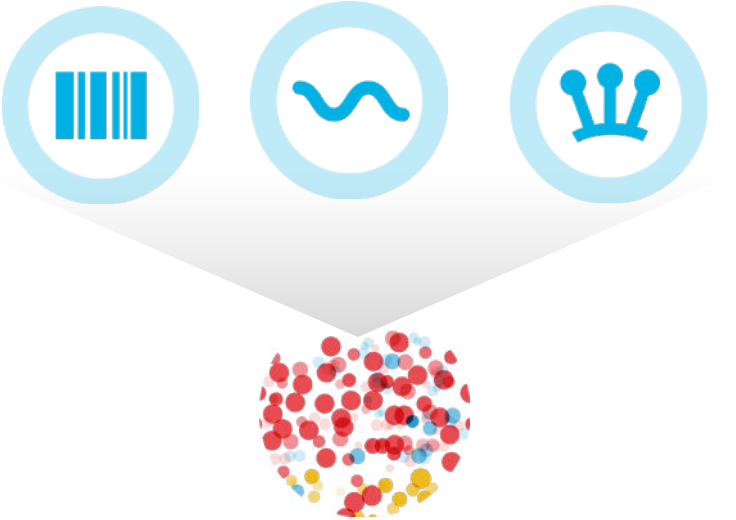


- Inputs:
  10x Gel Beads, Reagents and single cells in suspension

- Outputs:
  Digital gene expression and cell surface protein expression or CRISPR perturbation profiles from every partitioned cell

Single Cell CNV

Single Cell ATAC



Single Cell Gene Expression & Immune Profiling

Challenging — Most challenging

gDNA

gDNA (in open chromatin)

mRNA

Suboptimal Integrity is a Leading Failure Phenotype

Accurate Cell Recovery
Goal: +/- 20% of target
Challenging sample: <80% of target
Root Causes:
*Sample integrity
*Technical cell counting

High library complexity (signal)
Goal: Cell type dependent
Challenging sample: Very low value
Root Causes:
*Sample integrity
*RNase

Low background (noise)
Goal >70%, Ideal >90%
Challenging sample: <70%
Root causes:
*Sample integrity
*Insufficient wash

Cell Ranger · neuron_10k_v3 · Cells from a combined cortex, hippocampus and subventricular zone of an E18 mouse

SUMMARY ANALYSIS

Estimated Number of Cells
11,843

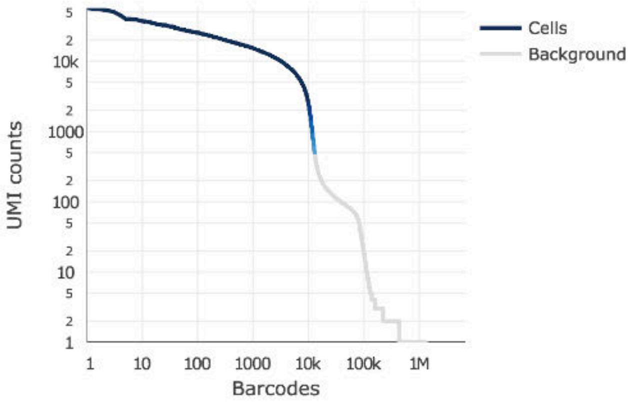
Mean Reads per Cell
30,153

Median Genes per Cell
2,679

Sequencing

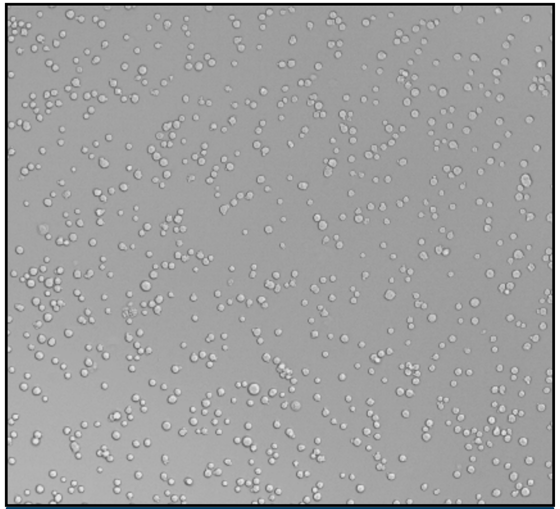
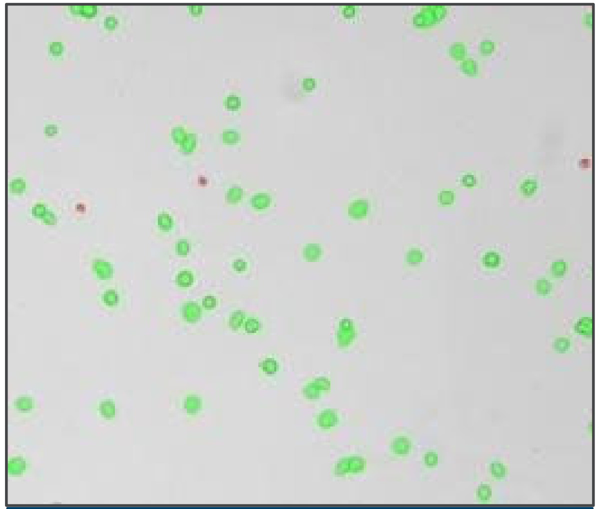
Number of Reads	357,111,595
Valid Barcodes	97.1%
Sequencing Saturation	49.2%
Q30 Bases in Barcode	97.1%
Q30 Bases in RNA Read	95.4%
Q30 Bases in Sample Index	93.8%
Q30 Bases in UMI	97.0%

Cells



Estimated Number of Cells	11,843
Fraction Reads in Cells	90.8%
Mean Reads per Cell	30,153
Median Genes per Cell	2,679
Total Genes Detected	21,080
Median UMI Counts per Cell	6,570

Quality is Critical

	Clean	Healthy	Intact
			
Goal is to minimize	• Aggregates/clumps • Subcellular debris • Free-floating RNA/DNA	• Biological decomposition • RNA leakage (background) • RNA degradation (signal)	• Physical decomposition • RNA leakage (background) • RNA degradation (signal)

General Cell Handling Recommendations

Debris/Aggregate Removal

- Use a cell strainer to remove aggregates or debris from washed cells
- The presence of cell aggregates, debris and/or fibers can result in inaccurate cell counts
- GEM generation occurs in microfluidic channels that are narrower than the typical human hair (i.e. < 100 µm) and the presence of cell debris or large aggregates may clog or wet the chip

Cell Counting

- Quantitate cells accurately before loading into the system
  - Approximately 65% loaded cells will be recovered
  - To maximize the likelihood of achieving the desired recovery target, the optimal input cell concentration is 700-1200 cells/µl
  - Recommended range: 500 to 10,000 recovered cells
  - Under- or over-loading may impact application performance

Storage of Single Cell Suspensions

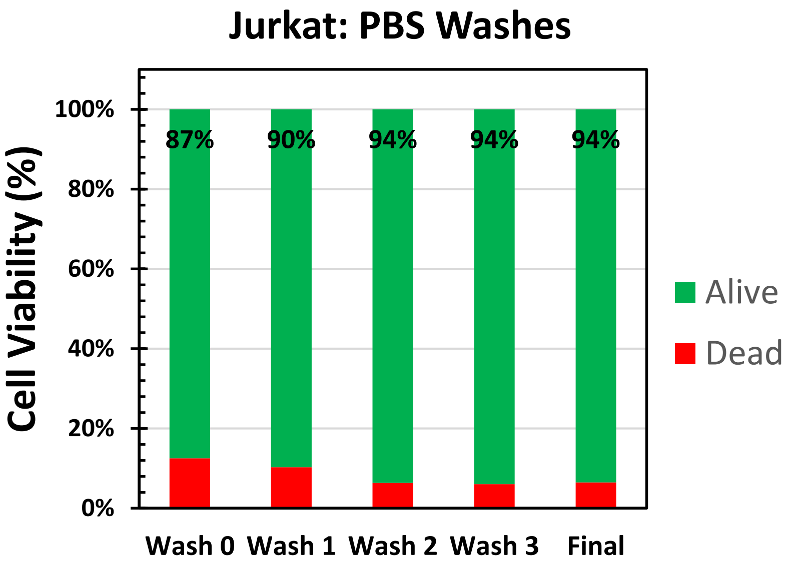
- Cell suspensions should always be kept on ice and where possible proceed with cell loading immediately after sample preparation
  - Ideally incubation time should be kept to a minimum (< 30 min)
- Some cell types are more fragile and cell viability may decrease significantly if not processed and loaded immediately

Cell Washing

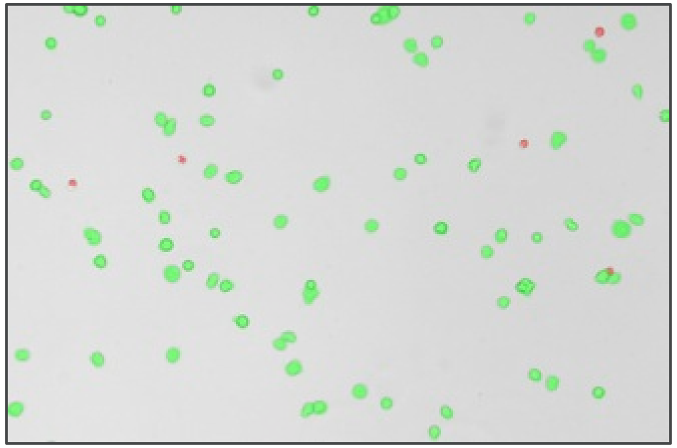
Washing isolated cells

- Transfer cells in media to a 2 mL Eppendorf tube
- Spin down cells to form pellet
    - Depending on cell size and concentration, pellet size varies
- Remove supernatant
- Gently add 1x PBS + 0.04% BSA away from cell pellet
- Gently pipette mix with Wide Bore pipette tip
- Repeat the wash one more time
- Spin down cells to form pellet
- Remove supernatant
- Resuspend cells in 1x PBS + 0.04% BSA with gentle pipette mix
    - **For accurate cell counting, do not invert tubes**
- Adjust to desired cell concentration

Note: PBS can be replaced with most common cell culture buffers and media if cells are unstable in PBS



Live/Dead Staining of Final Suspension



Spin Speed Optimized to Recover Live Cells

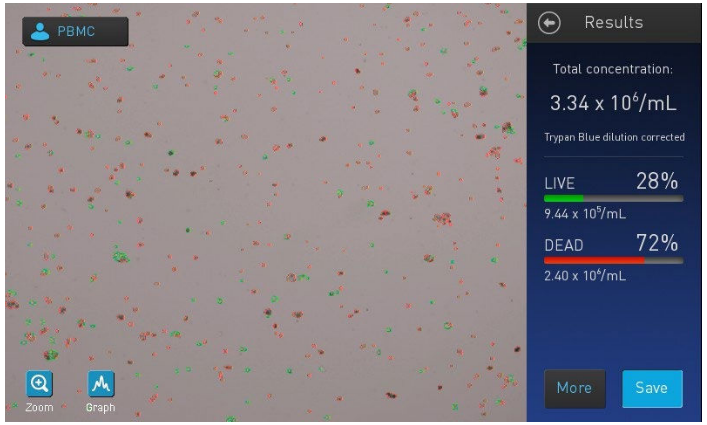
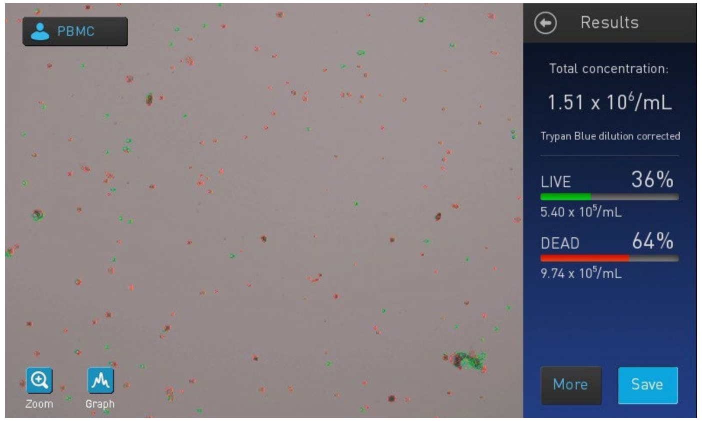
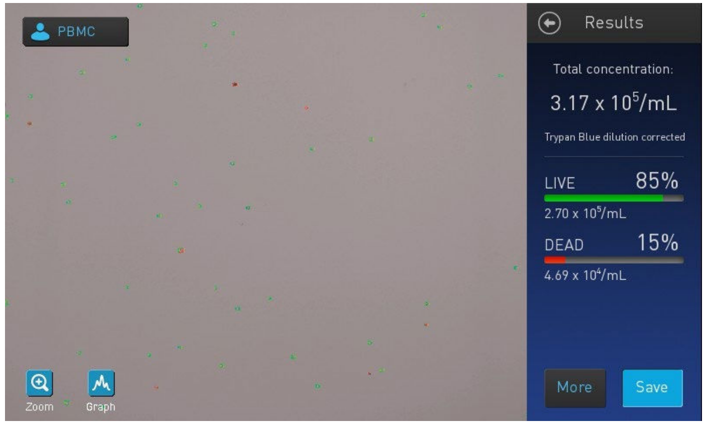
	Clear Cell Renal Carcinoma	Colorectal Cancer
Before		
After filtering and dead cell removal spins		

Image 1:

Image 2:

Image 3:

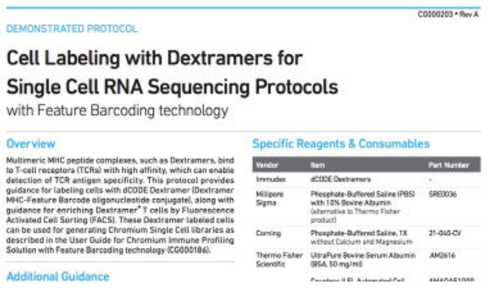
Image 4:

Alternative Buffer and Media

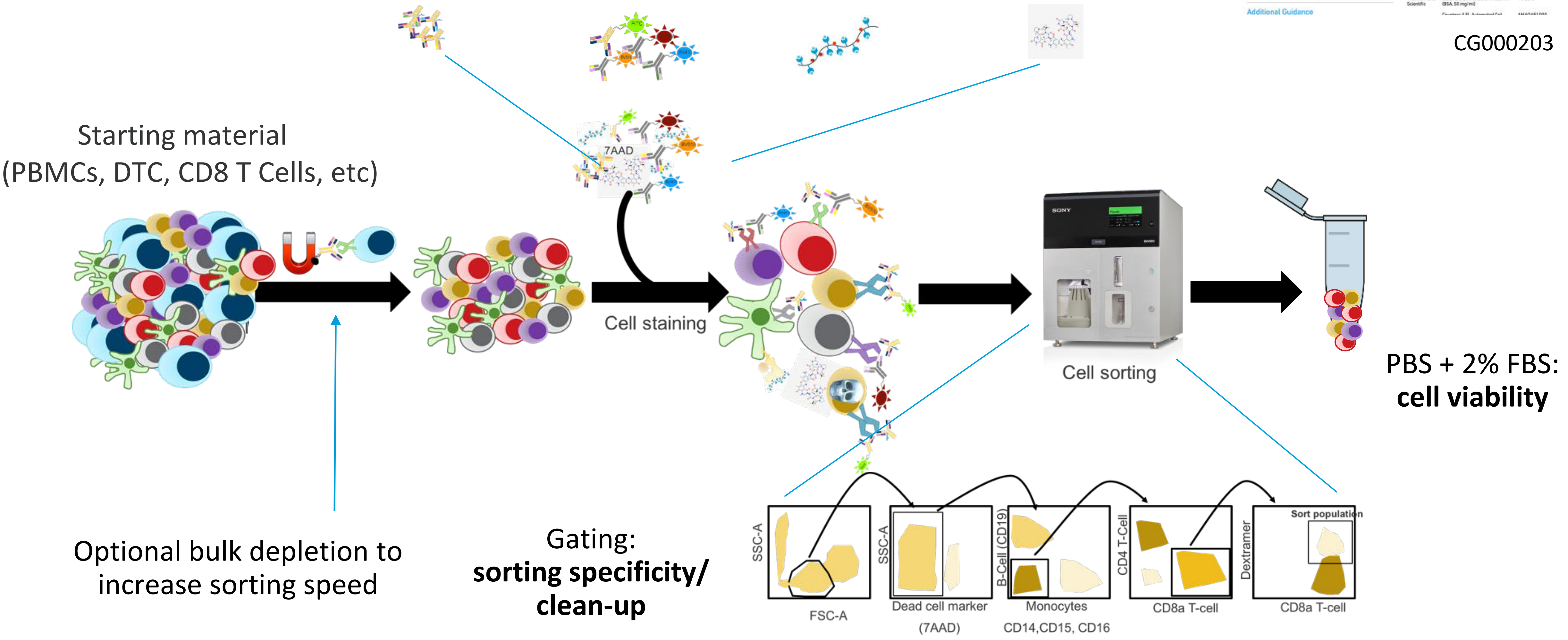
Tested in-house

- Tested input volume: 2.5 and 33 µl
- **Alternative Buffer**: no influence on performance
  - Dulbecco's Phosphate-Buffered Saline (DPBS)
  - Hank's Balanced Salt Solution (HBSS)
- **Alternative Media**: minimal reduction to no loss in performance
  - Eagle's Minimum Essential Medium (EMEM) + 10% FBS
  - Dulbecco's Modified Eagle Medium (DMEM) + 10% FBS
  - Iscove's Modified Eagle Medium (IMEM) + 10% FBS
  - Roswell Park Memorial Institute (RPMI) + 10% FBS
  - Ham's F12 + 10% FBS
  - 1:1 DMEM/F12 +10% FBS
  - M199

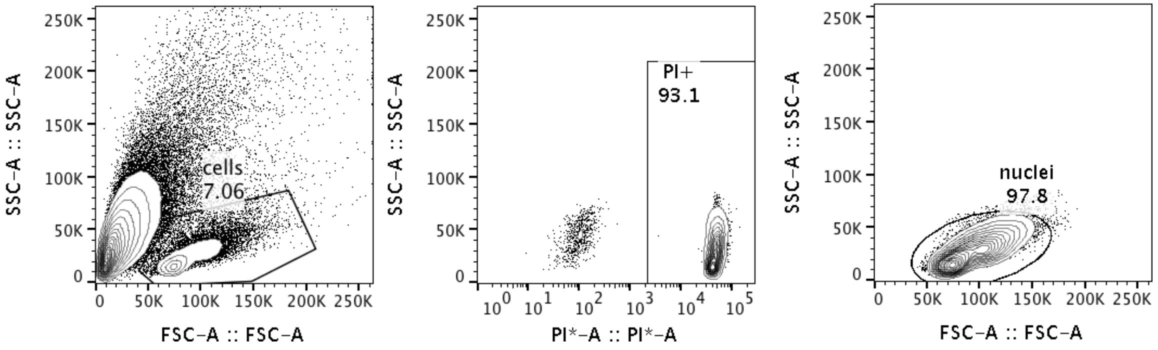
Sample Prep Workflow: Rare Cell Population



CG000203



Example: Frozen Tissue GEX



Protocol goal –
Maintain integrity by working quickly:

- Mechanical dissociation
- Use FACS to remove debris and ambient mRNA
- Collect nuclei in GEM-RT mix and process immediately

Why is this sample challenging?

- Small piece of tissue
- Frozen tissue

References

- 10x Genomics Customer Developed Protocol from Luciano Marletotto

Chromium Single Cell Gene Expression Solution with Feature Barcoding technology

Single-use microfluidics chip

Recovery well

Gel Bead well

Sample well

- Up to 8 channels processed in parallel
- 500 to 10000 cells per channel
- 18 minute run time (per chip)
- Up to 30 µm cell diameter tested
- Up to 65 % cell processing efficiency
- Temperature Range 18-28˚C

User controlled trade-off between cell numbers and doublet rate

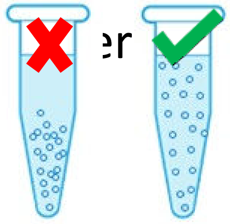
Number of Cells Loaded	Number of Cells	Expected Multiplet Rate (%)*
~800	~500	~0.4
~1,600	~1,000	~0.8
~8,000	~5,000	~3.9
~16,000	~10,000	~7.6

Firmware 4.00 or higher is required for running this assay

Single Cell Gene Expression Workflow

Determine Targeted Number of Cells

- Cell Suspension Volume Calculator Table is provided in the User Guide
- Targeted Cell Recovery numbers take into account a 65% cell processing efficiency
- For Example: Targeting 5000 cells with a cell concentration of 1000 cells/µl:
  - Add 35.0 µl water to aliquoted Master Mix
  - Add 8.3 µl cell suspension to Master Mix containing water
- Remember to resuspend the cells gently whe er taking aliquots to count or add to Master Mi



Next GEM

Cell Suspension Volume Calculator Table

(for step 1.2 of Chromium Next GEM Single Cell 3' v3.1 protocol)

Volume of Cell Suspension Stock per reaction (µl) | Volume of Nuclease-free Water per reaction (µl)

Cell Stock Concentration (Cells/µl)	Targeted Cell Recovery 500	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
100	8.3 / 35.0	16.5 / 26.7	33.0 / 10.2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
200	4.1 / 39.1	8.3 / 35.0	16.5 / 26.7	24.8 / 18.5	33.0 / 10.2	41.3 / 2.0	n/a	n/a	n/a	n/a	n/a
300	2.8 / 40.5	5.5 / 37.7	11.0 / 32.2	16.5 / 26.7	22.0 / 21.2	27.5 / 15.7	33.0 / 10.2	38.5 / 4.7	44.0 / -0.8	n/a	n/a
400	2.1 / 41.1	4.1 / 39.1	8.3 / 35.0	12.4 / 30.8	16.5 / 26.7	20.6 / 22.6	24.8 / 18.5	28.9 / 14.3	33.0 / 10.2	37.1 / 6.1	41.3 / 2.0
500	1.7 / 41.6	3.3 / 39.9	6.6 / 36.6	9.9 / 33.3	13.2 / 30.0	16.5 / 26.7	19.8 / 23.4	23.1 / 20.1	26.4 / 16.8	29.7 / 13.5	33.0 / 10.2
600	1.4 / 41.8	2.8 / 40.5	5.5 / 37.7	8.3 / 35.0	11.0 / 32.2	13.8 / 29.5	16.5 / 26.7	19.3 / 24.0	22.0 / 21.2	24.8 / 18.5	27.5 / 15.7
700	1.2 / 42.0	2.4 / 40.8	4.7 / 38.5	7.1 / 36.1	9.4 / 33.8	11.8 / 31.4	14.1 / 29.1	16.5 / 26.7	18.9 / 24.3	21.2 / 22.0	23.6 / 19.6
800	1.0 / 42.2	2.1 / 41.1	4.1 / 39.1	6.2 / 37.0	8.3 / 35.0	10.3 / 32.9	12.4 / 30.8	14.4 / 28.8	16.5 / 26.7	18.6 / 24.6	20.6 / 22.6
900	0.9 / 42.3	1.8 / 41.4	3.7 / 39.5	5.5 / 37.7	7.3 / 35.9	9.2 / 34.0	11.0 / 32.2	12.8 / 30.4	14.7 / 28.5	16.5 / 26.7	18.3 / 24.9
1000	0.8 / 42.4	1.7 / 41.6	3.3 / 39.9	5.0 / 38.3	6.6 / 36.6	8.3 / 35.0	9.9 / 33.3	11.6 / 31.7	13.2 / 30.0	14.9 / 28.4	16.5 / 26.7
1100	0.8 / 42.5	1.5 / 41.7	3.0 / 40.2	4.5 / 38.7	6.0 / 37.2	7.5 / 35.7	9.0 / 34.2	10.5 / 32.7	12.0 / 31.2	13.5 / 29.7	15.0 / 28.2
1200	0.7 / 42.5	1.4 / 41.8	2.8 / 40.5	4.1 / 39.1	5.5 / 37.7	6.9 / 36.3	8.3 / 35.0	9.6 / 33.6	11.0 / 32.2	12.4 / 30.8	13.8 / 29.5
1300	0.6 / 42.6	1.3 / 41.9	2.5 / 40.7	3.8 / 39.4	5.1 / 38.1	6.3 / 36.9	7.6 / 35.6	8.9 / 34.3	10.2 / 33.0	11.4 / 31.8	12.7 / 30.5
1400	0.6 / 42.6	1.2 / 42.0	2.4 / 40.8	3.5 / 39.7	4.7 / 38.5	5.9 / 37.3	7.1 / 36.1	8.3 / 35.0	9.4 / 33.8	10.6 / 32.6	11.8 / 31.4
1500	0.6 / 42.7	1.1 / 42.1	2.2 / 41.0	3.3 / 39.9	4.4 / 38.8	5.5 / 37.7	6.6 / 36.6	7.7 / 35.5	8.8 / 34.4	9.9 / 33.3	11.0 / 32.2
1600	0.5 / 42.7	1.0 / 42.2	2.1 / 41.1	3.1 / 40.1	4.1 / 39.1	5.2 / 38.0	6.2 / 37.0	7.2 / 36.0	8.3 / 35.0	9.3 / 33.9	10.3 / 32.9
1700	0.5 / 42.7	1.0 / 42.2	1.9 / 41.3	2.9 / 40.3	3.9 / 39.3	4.9 / 38.3	5.8 / 37.4	6.8 / 36.4	7.8 / 35.4	8.7 / 34.5	9.7 / 33.5
1800	0.5 / 42.7	0.9 / 42.3	1.8 / 41.4	2.8 / 40.5	3.7 / 39.5	4.6 / 38.6	5.5 / 37.7	6.4 / 36.8	7.3 / 35.9	8.3 / 35.0	9.2 / 34.0
1900	0.4 / 42.8	0.9 / 42.3	1.7 / 41.5	2.6 / 40.6	3.5 / 39.7	4.3 / 38.9	5.2 / 38.0	6.1 / 37.1	6.9 / 36.3	7.8 / 35.4	8.7 / 34.5
2000	0.4 / 42.8	0.8 / 42.4	1.7 / 41.6	2.5 / 40.7	3.3 / 39.9	4.1 / 39.1	5.0 / 38.3	5.8 / 37.4	6.6 / 36.6	7.4 / 35.8	8.3 / 35.0

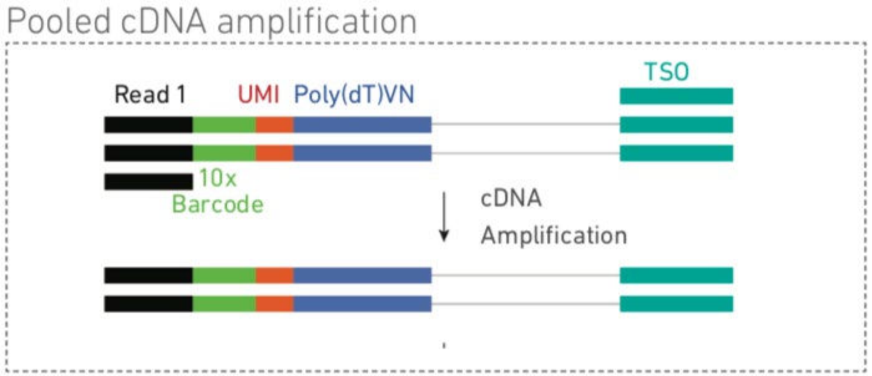
Single Cell Gene Expression Workflow

cDNA Amplification

- Prepare cDNA Amplification Reaction Mix
  - Primers must be selected based on whether generating only Gene Expression Libraries, or Gene Expression and Cell Surface Protein or CRISPR Libraries.
- Add 65 µl cDNA Amplification Reaction Mix to 35 µl of sample

Lid Temperature	Reaction Volume	Run Time
105°C	100 µl	~30-45 min
Step	**Temperature**	**Time**
1	98°C	00:03:00
2	98°C	00:00:15
3	63°C Version Specific Updated Temperature	00:00:20
4	72°C	00:01:00
5	Go to Step 2, see table below for total # of cycles	
6	72°C	00:01:00
7	4°C	Hold

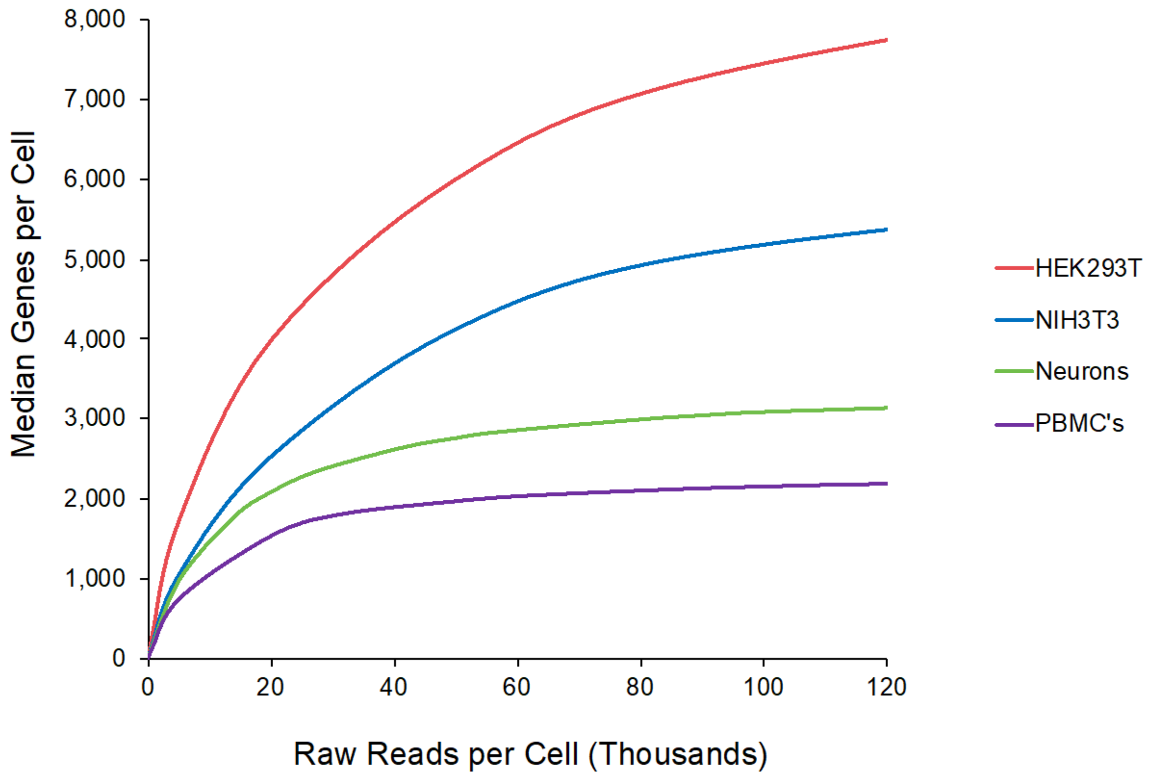
- Use Cell Load to determine number of Total Cycles for cDNA amplification



Cell Load	Total Cycles
<500	13
500–6,000	12
>6,000	11

Sequencing Depth for Typical Samples

Gene Expression Libraries

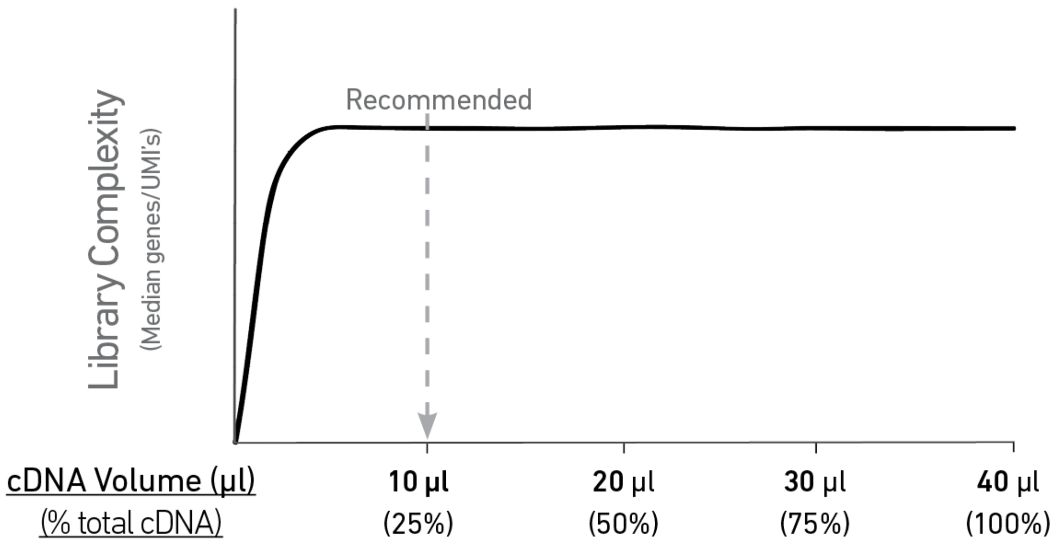


- 20,000 raw read pairs per cell is the recommended minimum sequencing depth for typical samples.
- Given variability in cell counting and loading, extra sequencing may be required if the cell count is higher than anticipated.

Single Cell Gene Expression Workflow

Carrying forward only 25% of Total cDNA is Sufficient to Capture Full Library Complexity

Correlation: cDNA Input vs Library Complexity

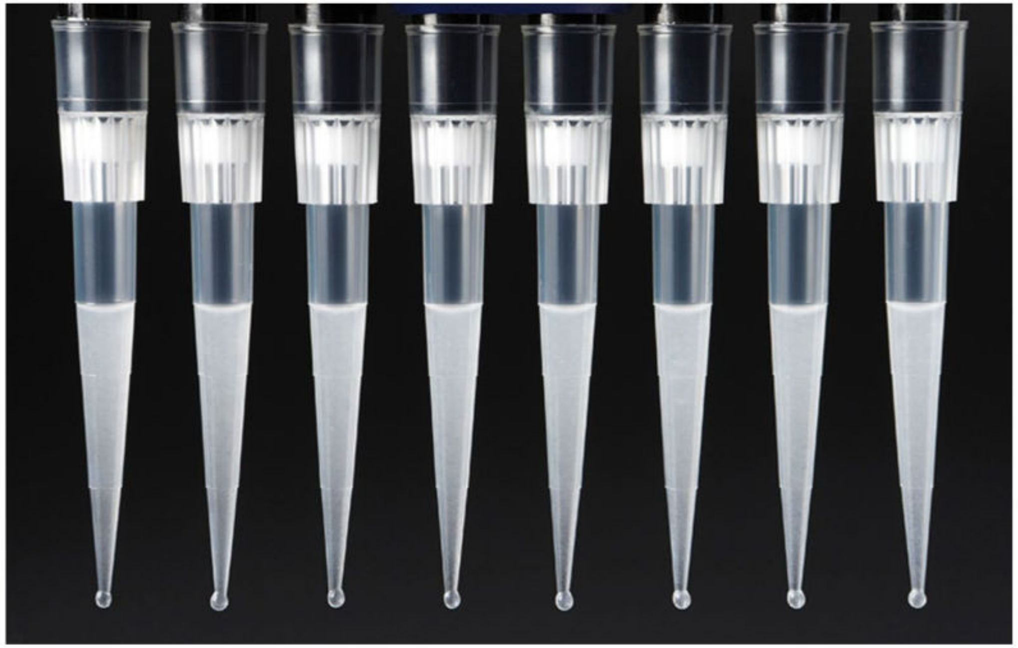


Library Construction Input Mass and SI PCR Cycle Examples

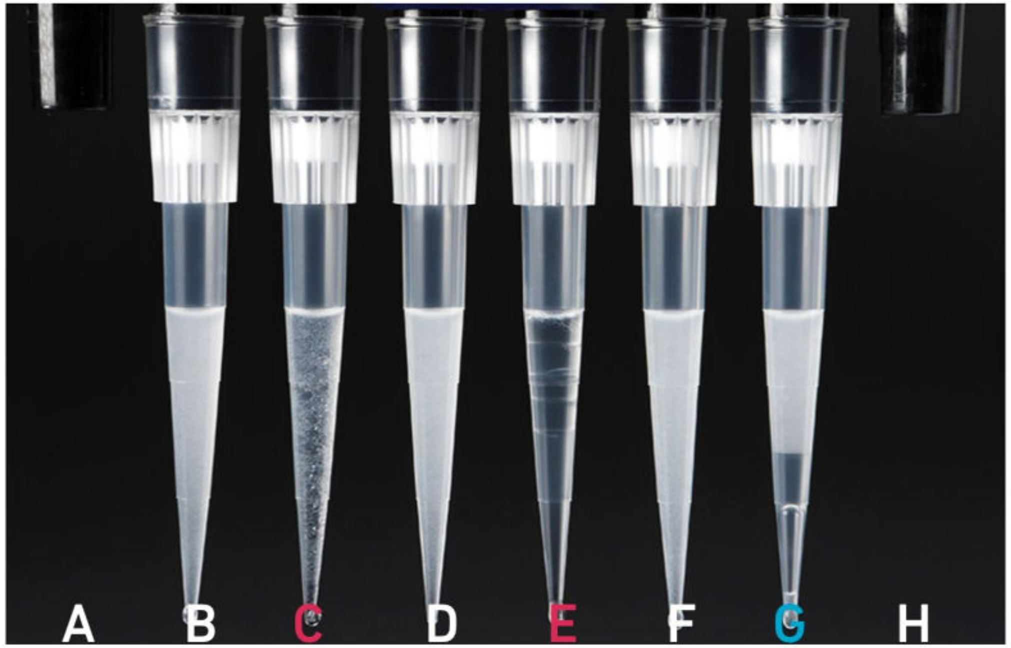
Cell Type	Targeted Cell Recovery	Total cDNA Yield (ng)	cDNA Input into Fragmentation		SI PCR Cycle Number
			Volume (µl)	Mass (ng)	
High RNA Content	Low	250 ng	10 µl	62.5 ng	13
	High	1900 ng	10 µl	475 ng	10
Low RNA Content	Low	1 ng	10 µl	0.25 ng	16
	High	200 ng	10 µl	50 ng	12

Single Cell Gene Expression Workflow

Visually Inspect Gems to Confirm the Presence of a Uniform Emulsion



All liquid levels are similar in volume and opacity without air trapped in the pipette tips.



Pipette tips C and E indicate a wetting failure. Pipette tip C contains partially emulsified GEMs. Emulsion is absent in pipette tip E. Pipette tip G indicates a reagent clog.

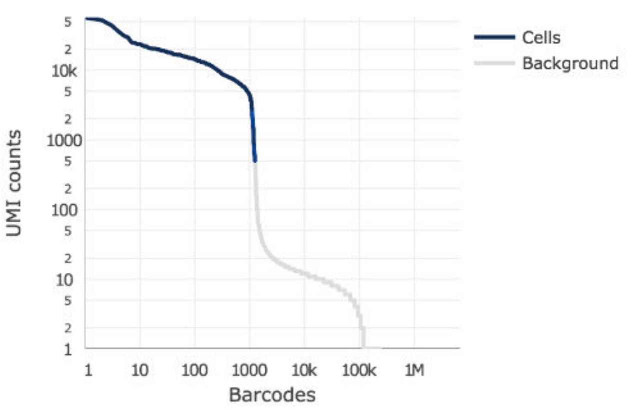
QC Metrics: Cell Calling

Wetting Failure

- A comparison of cell calling between a typical sample and a sample with wetting failure.
- Estimated number of cells is unreliable in such failure modes.

Typical Sample

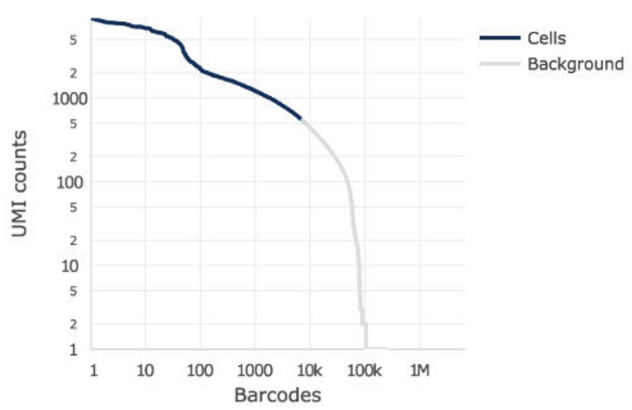
Cells



Estimated Number of Cells	1,222
Fraction Reads in Cells	94.9%
Mean Reads per Cell	54,502
Median Genes per Cell	1,919
Total Genes Detected	18,391
Median UMI Counts per Cell	6,628

Loss of Single Cell Behavior (eg. Wetting failure)

Cells



Estimated Number of Cells	7,307
Fraction Reads in Cells	37.5%
Mean Reads per Cell	4,677
Median Genes per Cell	436
Total Genes Detected	15,426
Median UMI Counts per Cell	762

Algorithm has trouble discerning cells from the background

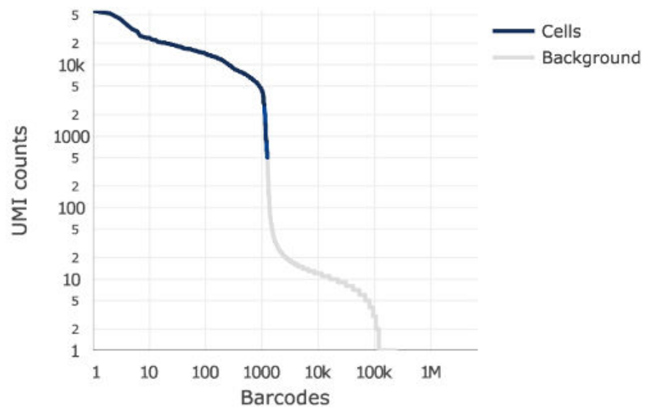
QC Metrics: Cell Calling

Clog

- A comparison of cell calling between a typical sample and a sample with a clog.
- Similar barcode rank profile can be seen for example for samples with clogs, low sequencing depth.
- The number of cells recovered will likely be lower than the targeted cell numbers. But the data for the cells called is usable.

Typical Sample

Cells



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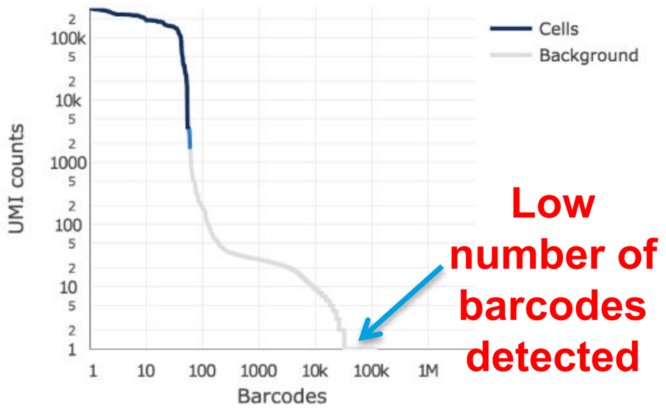
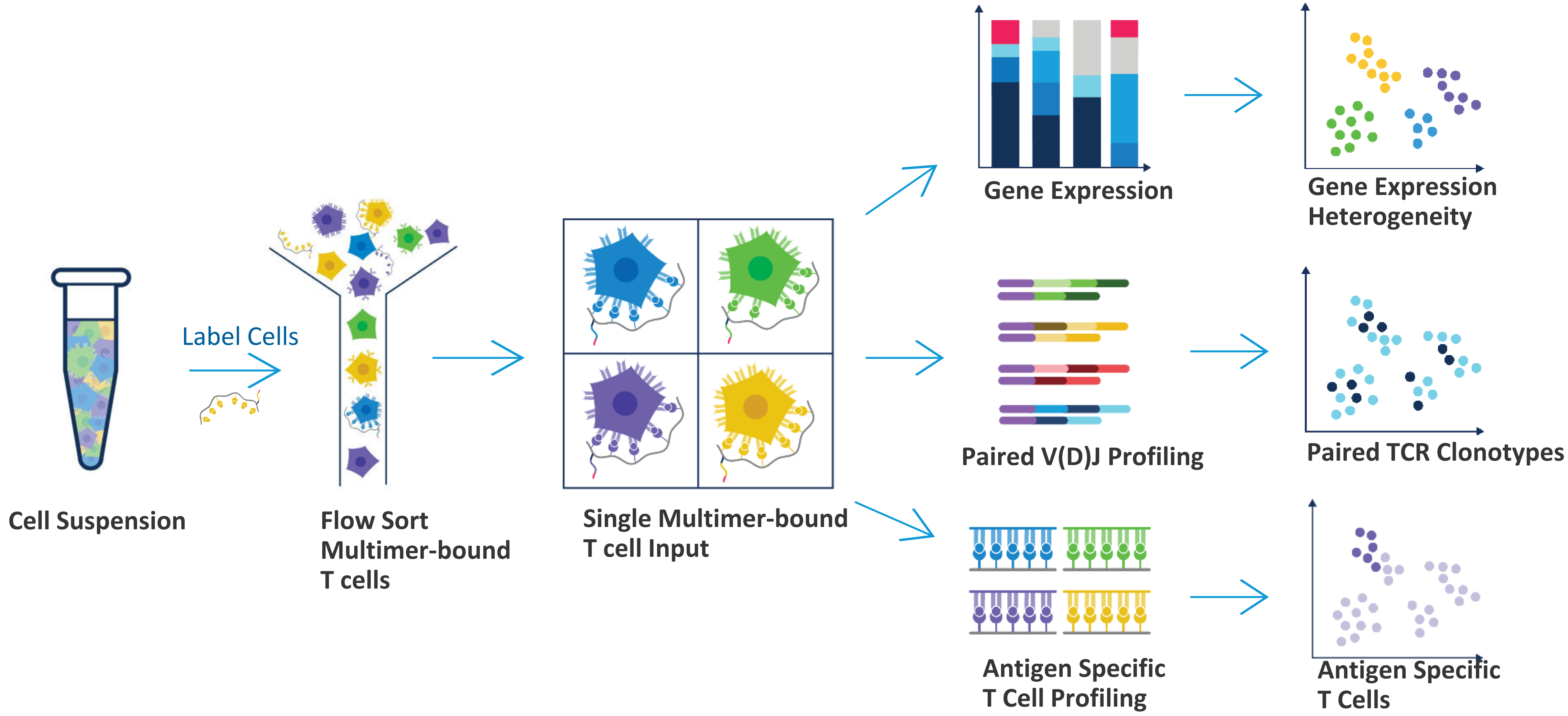
Low Barcode counts (eg. Clog, low-depth)

Cells



Estimated Number of Cells	59
Fraction Reads in Cells	97.1%
Mean Reads per Cell	847,584
Median Genes per Cell	8,936
Total Genes Detected	15,883
Median UMI Counts per Cell	151,578

Reveal Antigen Specificity with Feature Barcoding Technology



Antibody Preparation

Custom Conjugation

Specific Reagents and Consumables

Vendor	Item	Part Number
Expedeon	ThunderLink Plus	425-0300
IDT	Feature Barcode Oligonucleotide (see Table 1 & Appendix)	-
	100 µg Purified Azide-free Antibody	

To perform custom conjugation, a customer needs to acquire:

- ThunderLink Plus kit
- Feature Barcode Oligonucleotide
  - 10 nmoles, HPLC purified, lyophilized
  - **NOT** resuspended in TE buffer
  - The BC structure must be compatible with desired assay
  - The BC sequence should be selected from the provided whitelist
    - This ensures compatibility if pooled with commercially available Feature Barcoding technology compatible antibodies
- 100 µg Antibody

10x Genomics Protocol	Feature Barcode Oligonucleotide Sequence
Single Cell 3' v3 – Cell Surface Protein (CG000185)	/5AmMC12/GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTNNNNNNNNNN-NNNNNNNNNNNNNNN-NNNNNNNNNGCTTTAAGGCCGGTCCTAGCAA TruSeq Read 2 · 10 nt · Feature Barcode (15 nt) · 9 nt · Capture Sequence 1
Single Cell V(D)J – Cell Surface Protein (CG000186)	/5AmMC12/CGGAGATGTGTATAAGAGACAGNNNNNNNNNN-NNNNNNNNNNNNNNN-NNNNNNNNNCCCATATAAGAAA Nextera partial Read 2 · 10 nt · Feature Barcode (15 nt) · 9 nt · Capture Sequence

Note: Consult Barcode Whitelist for Custom Feature Barcoding conjugates (Document CG000193), for more information

Dissociation Example: Single Cell ATAC-seq

Fresh mouse breast

Mince with scalpel blade on ice

Lysis buffer (1x detergent) | Lysis buffer (0.1x detergent)

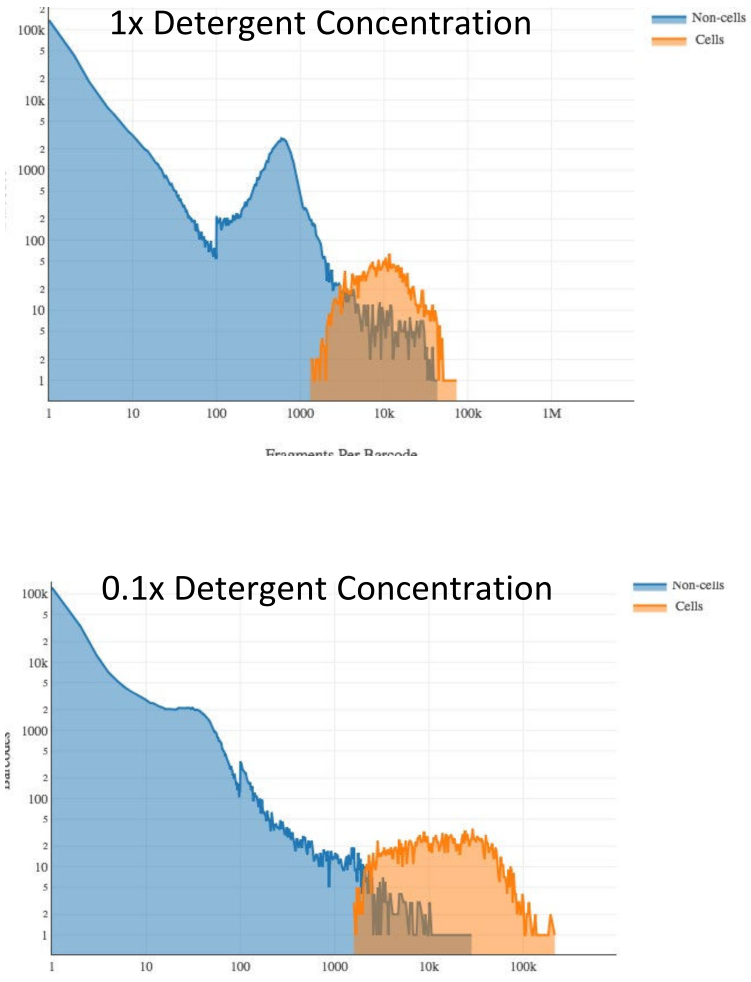
10 min on ice

Spin 5 min 500 rcf

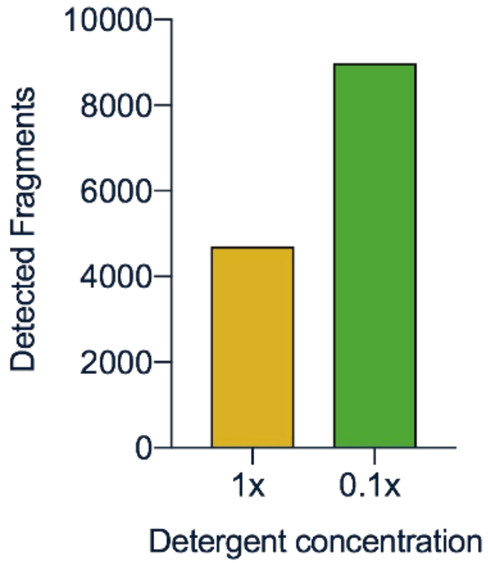
Wash

Resuspend in diluted nuclei buffer

Signal-to-noise

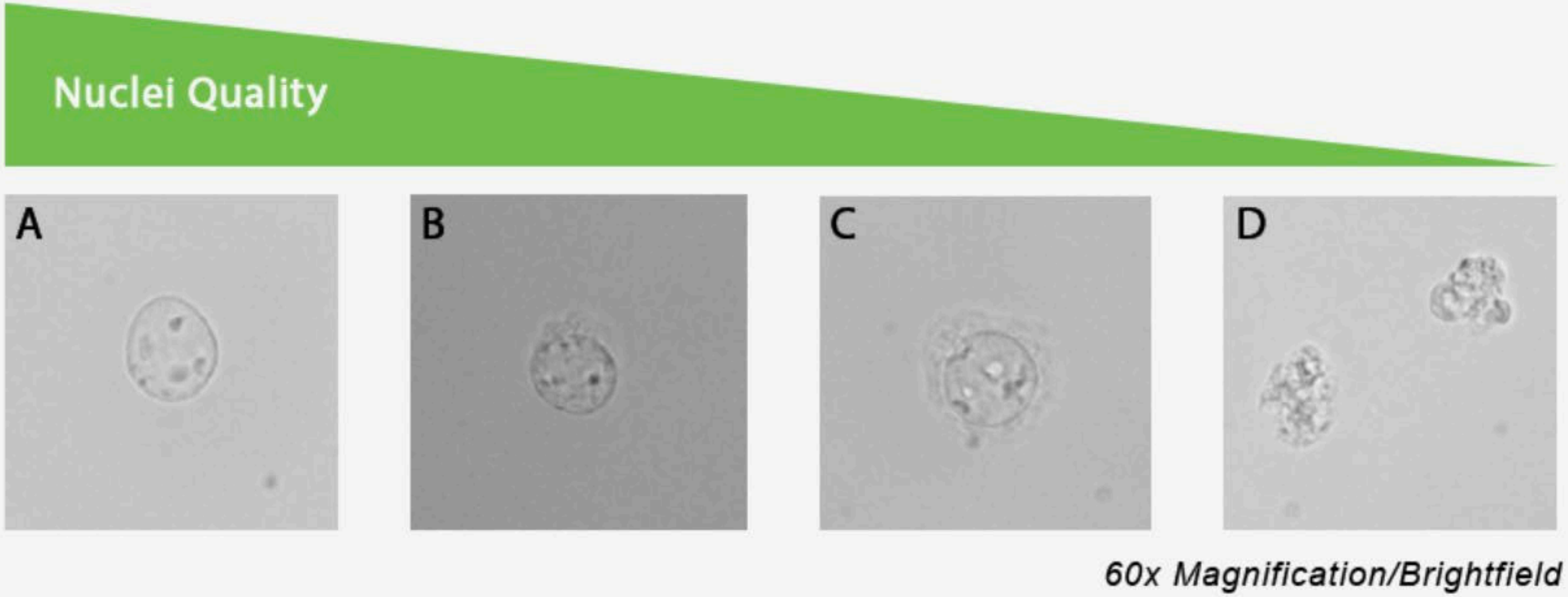


Sensitivity



Results: **Gentler lysis** produced higher quality ATAC data

Sample Quality Assessment



A: High-quality nuclei have well-resolved edges. Optimal quality for single cell ATAC libraries.

B: Mostly intact nuclei with minor evidence of blebbing. Quality single cell ATAC libraries can still be produced.

C: Nuclei with strong evidence of blebbing. ***Proceed at your own risk.***

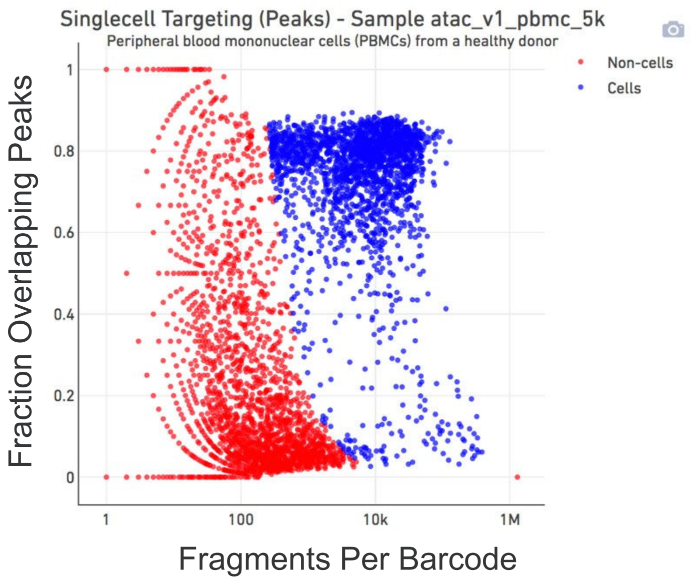
D: Nuclei are no longer intact. ***Do not proceed!***

Filter Barcodes

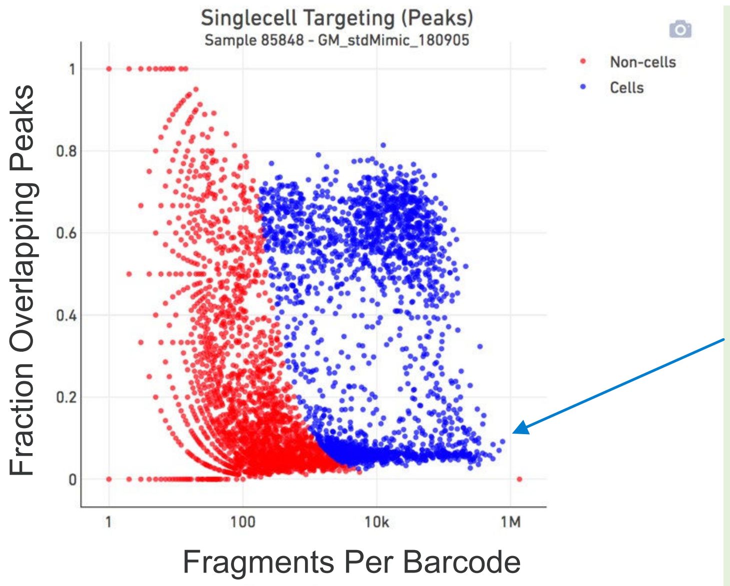
- Align
- Count
- Smooth
- Threshold
- Call Peaks
- **Filter**
- Call Cells
- Matrix

Bulk Analysis

- Some barcodes have very few *overlaps with called peaks*. Should they be kept?



Ideal library



- High number of neutrophils and dead cells
- Low in targeting/ ATAC peak signal/ TSS score
- Insert size distribution is enriched in mono-nucleosome fragments, no DNA twist

Summary of Key Lessons

When our standard guidance isn't applicable:

- Treat cells gently and minimize decomposition
  - Use gentl(er) lysis conditions
  - Reduce wash steps*
  - Use a swinging bucket centrifuge
  - Keep cells in media + FBS instead of PBS**
- Work quickly
  - Consider sorting, it is a versatile tool for sample prep
  - Minimize unnecessary handling steps
- *Consider the benefits and drawbacks of every different technique*

*Cell surface protein analysis requires thorough washing
**ATAC has specific buffer formulation

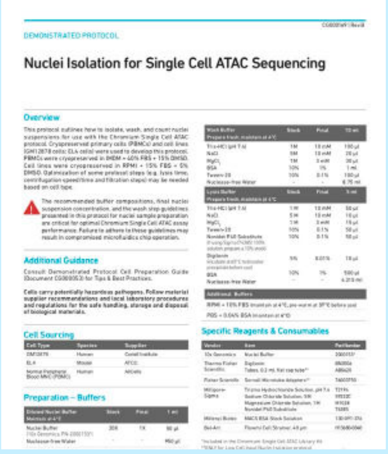
Sample Prep Support

10x-pert Webinars

High level overviews of Sample Prep

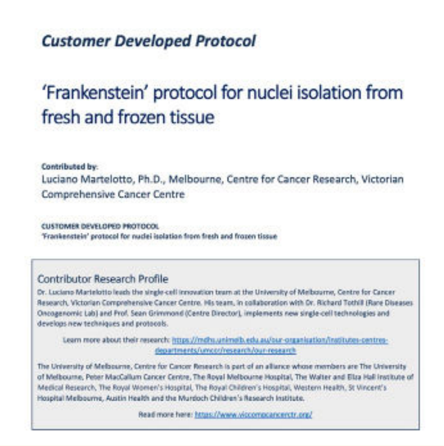
Demonstrated Protocols & Technical Notes

Learn foundational techniques



Customer Developed Protocols & Publications

Find specific examples



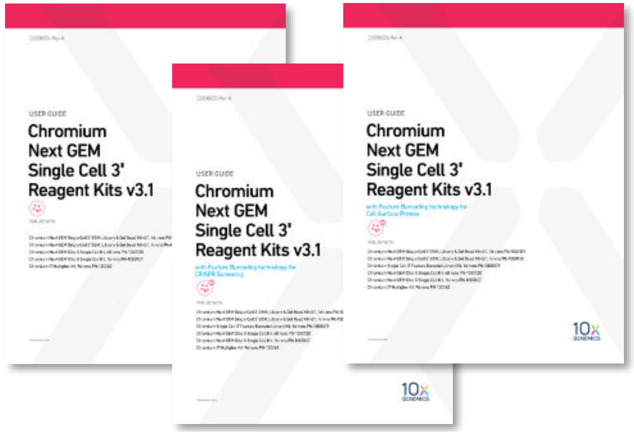
World-Class Support Team

Talk to the experts

Resources

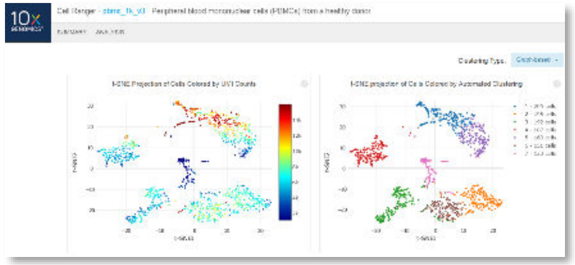
Demonstrated Protocols

User Guides



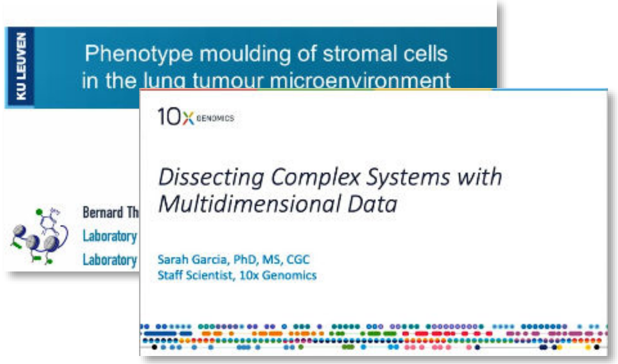
Application Notes

Public Data Sets



How-To Videos

Scientific Seminars



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